# Long-range information and physicality constraints improve predicted protein contact maps

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Protein topology representations such as residue contact maps are an important in-

termediate step towards *ab initio* prediction of protein structure, but the problem of predicting reliable contact maps is far from solved. One of the main pitfalls of existing contact map predictors is that they generally predict unphysical maps, i.e. maps that cannot be embedded into three-dimensional structures, or, at best, violate a number of basic constraints observed in real protein structures, such as the maximum number of contacts for a residue.

Here we focus on the problem of learning to predict more "physical" contact maps. We do so by first predicting contact maps through a traditional system (XXStout), then filtering these maps by an ensemble of artificial neural networks. The filter is provided as input not only the bare predicted map, but also a number of global, or long-range features extracted from it.

In a rigorous cross-validation test, we show that the filter greatly improves the predicted maps it is input. CASP7 results, on which we report here, corroborate this finding. Importantly, since the approach we present here is fully modular, it may be beneficial to any other *ab initio* contact map predictor.

*Keywords*: protein structure prediction; contact maps; neural networks.

## 1. Introduction

Protein topology representations such as residue contact maps are an important intermediate step towards *ab initio* prediction of protein structure. In fact, it has been argued that, for an adequate definition of contact, a contact map is roughly equivalent to the structure itself [1].

Not surprisingly, the problem of accurately predicting residue contact maps from primary sequences is still largely unsolved. Among the reasons for this are the unbalanced nature of the problem (with far fewer examples of contacts than non-contacts, although this depends on the definition of contact) and, especially, the formidable challenge of capturing long-range interactions in the maps.

In order to mitigate the intrinsic difficulty of mapping one-dimensional input sequences into two-dimensional outputs, in virtually all existing predictive systems protein contact maps are inferred by modelling a set of independent tasks, each task being the prediction of whether two residues are in contact[2,3,4,5,6].

Although the resulting problem turns out to be simplified, this approach does not take into account the global nature of the task. For this reason, joining the predictions for all residue pairs generally corresponds to a map that is not physically realisable. This means that the structure derived from this contact map violates basic constraints observed in real protein structures (e.g. the maximum number of contacts per amino acid) or, in the worst case, that the residues cannot be embedded into the three dimensional (3D) euclidean space.

Here we focus on the problem of learning to predict physically realisable residue contact maps. How to directly incorporate rules encoding protein structural principles into a learning framework remains an open issue. In this article we present the results of a straightforward approach: we predict contact maps by a traditional algorithm[6] and then train an ensemble of artificial neural networks (NN-Filter) to filter these maps, with the aim of correcting some of the errors they contain and of increasing their physicality. This NN-Filter is explicitly provided as input not only

the bare predicted map, but also information about its physical realisability and compressed long-range contact information extracted from it.

Predicted contact maps are obtained from XXStout[6] that uses as input evolutionary information from alignments of multiple homologous sequences, predicted secondary structure[7], solvent accessibility and contact density [6,8]. XXStout's predictions contain errors clearly due to to the fact that each pair of amino acids is modelled as a separate instance of the problem. For instance patterns of contacts between secondary structure elements, especially for large sequence separations, are shaped differently from those found in real contact maps (they tend to come in squares - see figures 11-12 for examples). Moreover amino acids are often predicted to be in contact with more or fewer other amino acids than is ever observed in native maps.

We implement the NN-Filter stage by training two different ensembles of artificial neural networks to process different regions of the contact map: the region close to the diagonal of the map, corresponding to small sequence separations; the region further away from the diagonal. We do so because the rules governing contact probability are likely to be different for positions near the main diagonal (mainly made by backbone atoms, reflecting secondary structure contacts) than for positions away from it (where contacts mainly occur between the side chains of amino acids placed in different secondary structure elements).

The NN-Filter stage is input a combination of local information about the predicted map -such as estimated probabilities of contact for pairs of residues, or secondary structure predictions for individual residues[7]- and global information, such as total number of contacts predicted for a residue or pair of residues, residue contact order, etc.

We perform experiments to validate the ability of NN-Filter to recover correct contact information and to model maps with higher chance of physical realisability. To measure this, beside standard performance measures like Precision, Recall and their harmonic mean $F_1$, we compare a number of distributions in real, predicted and filtered contact maps. The distributions considered are that of the number of contacts per residue, of the number of contacts with both residues in each pair in contact, and of the contact order per residue.

In a rigorous cross-validation experiment we show that filtered maps are a significant improvement over first stage predictions. We submitted both XXStout and NN-Fiter predictions to the CASP7 competition. Here we report our CASP7 results, which confirm that NN-Filter greatly improves on "raw" XXstout predictions.

Since the NN-Filter stage is entirely modular, we believe that our approach is likely to be beneficial to any contact map predictor, provided that the filter is trained on its error distribution.

| set name | used to |
|---|---|
| $TRS1280$ | choose NN-Filter parameters |
| $VS321$ | choose NN-Filter hyper-parameters |
| $TES481$ | evaluate NN-Filter performances |
| $TES258$ | evaluate 3D reconstruction performances |

Table 1. Data sets used in this work. See text for details.

## 2. Methods

### 2.1. *Data*

#### 2.1.1. *Datasets*

The training dataset used in this study was extracted from the December 2003 25% PDBSELECT list [9]. First, sequences longer than 200 amino acids are excluded, leaving a total of 1601 sequences with 163376 residues (set $PDB1601$). One every five sequences is then included into the validation set, which we use to select the hyperparameters of the systems, such as the architectural details of the neural networks, the number of training epochs to be performed, etc. The validation set contains 321 sequences and 32725 residues and we will refer to it as $VS321$. The remaining sequences (1280 in total - we will refer to the overall set as $TRS1280$) are adopted as a training set.

We create a further set ($TES481$) to assess the quality of filtered maps vs "raw" maps in an unbiased manner. From the July 2005 25% PDBSELECT list we remove all those sequences with sequence identity greater than 25% with any of the sequences found in $VS321$, $TRS1280$ and $TES258$ (see below). Of the remaining sequences we remove those that have amino acids with no 3D coordinates, are longer than 200 residues. This set contains 481 sequences (49159 amino acids).

Finally, we adopt a further set of 258 proteins (33667 residues - $TES258$) of maximal length 200 to test if NN-Filter maps yield better 3D reconstructions than "raw" XXStout ones. The set, also described in [10,11], is composed of sequences with no sequence identity greater than 25% among them, and no sequence identity greater than 25% against any sequence in the 2003 25% PDBSELECT list, hence no sequence identity greater than 25% against any sequence in $VS321$ or $TRS1280$.

There are two main reasons for using only chains up to 200 amino acids in length: to restrict our analysis mostly to proteins containing single domains; because XXstout[6], which is the source of "raw" maps, is trained on proteins of 200 residues at most.

Overall, all sets used in this work ($TRS1280$, $VS321$, $TES481$, $TES258$) have a maximal sequence identity of 25% both internally and with respect to the other sets, i.e. no two sequences used in this work show more than 25% identity between them, no matter what sets they are from. Table 1 summarises the roles of the various data sets.

2.1.2. *Multiple sequence alignments*

For all proteins in all sets we created multiple sequence alignments from NR database as available on March 2004, redundancy reduced at a 98% threshold (1.05 million sequences). The alignments are generated by three runs of PSI-BLAST [12] with parameters b = 3000, e = $10^{-3}$ and h = $10^{-10}$.

2.1.3. *Contact Maps*

The contact map of a protein with $N$ amino acids is a symmetric $N \times N$ matrix $C$, with elements $C_{ij}$ defined as:

$$C_{ij} = \begin{cases} 1 & \text{if amino acid i and j are in contact} \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

We define two amino acids as being in contact if the euclidean distance between their $C_\alpha$ is less than a given threshold. Here we use 12Å and 8Å thresholds.

Positions beside the diagonal (the diagonal being all $C_{ij}$ for which $i = j$) likely follow different rules than those away from it, those contacts close to the diagonal reflecting secondary structure/local interactions, with those further away reflecting long range interactions mainly between different secondary structure elements (see [13]). For this reason we train two different sets of neural networks, one to filter map positions beside the diagonal and one for positions away from it. The cutoff between "beside" and "away" contacts is chosen as follows: we check sequence separations for all contacts between amino acids that are part of the same secondary structure element in all the proteins belonging to December 2003 25% PDBSELECT list; we increase the highest margin found (10 for 12Å and 6 for 8Å) by 2, yielding cutoffs of 12 and 8 for 12Å and 8Å respectively. As the $C_\alpha$ of an amino acid is always closer than 8Å to the $C_\alpha$ of the neighbouring ones in the sequence, map postions just beside diagonal are set to contact by default.

### 2.2. *XXStout*

Predicted contacts are obtained from contact map prediction system XXStout[6]. XXStout predicts contacts at three different thresholds: 6Å, 8Å and 12Å. In this work we use 8Å and 12Å maps. Contact maps are predicted as follows: protein secondary structure, solvent accessibility and contact density are predicted from the sequence using, respectively, Porter[7,8], PaleAle[8] and BrownAle[6]; ensembles of two-dimensional Recursive Neural Networks [14,6] predict the contact maps based on the sequence, a profile of amino-acid frequencies obtained from multiple alignments of the sequence against the NR, and predicted secondary structure, solvent accessibility and contact density. XXStout is trained and tested in cross-validation on the $PDB1601$ set, i.e. the sum of training and validation set for NN-Filter (respectively $TRS1280$ and $VS321$). For this reason, any improvement by NN-Filter over XXStout cannot be attributed to differences in training set size or coverage.

### 2.3. *Artificial Neural Networks*

We use fully connected, feed forward neural networks trained via the backpropagation algorithm by minimising the cross-entropy error between the output and target probability distributions. We use a hybrid between batch and online training, in which network weights are updated more than once during a training epoch. Training examples are randomly shuffled before each training epoch. We set the number of hidden neurons to 20 after preliminary experiments and do not change it thereafter. Training is always stopped after 2000 epochs.

Due to the large number of examples and the long time needed to train a single network on all training instances, the training set is further divided into 20 sets (away from diagonal) and 5 sets (beside diagonal), each containing approximately the same number of examples. A different network is trained on each of these subsets and then the networks are joint in an ensemble of 20 for map positions away from the diagonal, 5 for map positions beside the diagonal.

### 2.4. *Input representation*

#### 2.4.1. *Local inputs*

The main input to the filtering neural networks is the "raw" map predicted by XXstout. We input a whole patch of 11x11 contact predictions. Thus a network that predicts the probability of contact between residues in positions $i$ and $j$ along the sequence is shown contact predictions (in the form of predicted probabilities of contact) for all pairs of amino acids $(k, l)$ with $i-5 \leq k \leq i+5$ and $j-5 \leq l \leq j+5$, for a total of 121 inputs. We also input Porter[7] secondary structure predictions for all 22 residues in the windows above, in the form of predicted probabilities of being in one of the 3 standard secondary structure classes (Helix, Strand, Coil). It should be noted that Porter is trained on the December 2003 25% PDBSELECT list, no sequence of which shows more than 25% sequence identity to any sequences in the sets we use for testing here ($TES258$ and $TES481$), hence the use of Porter does not violate the separation between training, validation and test sets. Sequence and evolutionary information are also provided to the network in the form of: identities of residues in positions $i-1$, $i$, $i+1$ and $j-1$, $j$, $j+1$; frequency profiles for residues in positions $i-1$, $i$, $i+1$ and $j-1$, $j$, $j+1$, extracted from multiple sequences alignments.

#### 2.4.2. *Global inputs*

The groups of inputs above provide local information about the map. We also adopt a number of input features that contain global information about a map:

- Binary 8Å and 12Å map values for $(i, j)$ obtained as follows: all the probabilities of contact, as predicted by XXStout, for pairs of amino acids at the same sequence separation ($|i-j|$) are added up yielding $\mathcal{L}$, then the $L$

top-scoring pairs at the given separation are classified as contacts, where $L$ is the integer part of $\mathcal{L}$. Notice that because of its definition, each element of these maps implicitly contains information about all those elements on the map with sequence separation $|i-j|$. This definition of contact is also used to determine the following features.

- Number of contacts per amino acid (NC) of amino acids $i$ and $j$.
- Number of amino acids in contact with both $i$ and $j$ (NCIJ).
- Residue Contact Order (CO): for each amino acid $i$ its CO ($CO_i$) is computed using equation 2, where $d_{ij}$ is the (euclidean) distance between amino acids $i$ and $j$, $t$ is the contact threshold, $L$ is the protein length and $NC_i$ is the total number of contacts for $i$.

$$CO_i = \frac{\sum_{\forall j: d_{ij} < t} |i-j|}{L \cdot NC_i} \tag{2}$$

- Sequence separation between amino acids $i$ and $j$: $|i-j|$.
- Protein length.

### 2.5. *Quality measures*

#### 2.5.1. *Standard measures*

Standard methods used to measure predicted map quality take into account correctly or incorrectly classified map positions (true positives, TP; false positives, FP; true negatives, TN; and false negatives, FN), that are used to compute Recall (R, equation 3), Precision (P, equation 4) and their harmonic average, F1 (equation 5), as follows:

$$R = \frac{TP}{(TP+FN)} \tag{3}$$

$$P = \frac{TP}{TP+FP} \tag{4}$$

$$F_1 = \frac{2PR}{P+R} \tag{5}$$

Both XXstout and NN-Filter output, for each pair of residues $(i, j)$, a number between 0 and 1 representing the estimated probability of contact, given the training examples observed. To match these values into "hard" predictions, a threshold needs to be established above which a network output is considered a contact and below which it is not. A ROC curve is obtained by testing precision $P$ and recall $R$ of a predictor for a sample of threshold values and plotting $R$ against $1-P$. We present ROC curves for all systems tested, and compute performances at breakeven points, i.e. those points on the ROC curve at which $R = P$. A further measure we report is the total number of contacts predicted and compare it in filtered and "raw" maps vs. native maps.

2.5.2. *Physicality measures*

Maps physicality (if the map represents a truly realisable 3D structure) is not taken into account by the quality measures above. To measure this, we run Kolmogorov-Smirnov tests on the distributions of a number of features related to physicality, against the distribution in native maps. We run the tests on filtered maps vs. native maps, and "raw" maps vs. native to test whether filtering improves the overall physicality of the maps. Features whose distribution we compare are:

- Number of contacts per amino acid (NC)
- Number of amino acids in contact with 2 amino acids already in contact (NCIJ)
- Residue Contact Order (CO)

## 3. Results

### 3.1. *ROC curves*

In figures 1, 2, 3 and 4, we present ROC curves on the $TES481$ set for XXstout and NN-Filter predictions, both for near-diagonal and long-range contacts, and for both contact definitions adopted (8Å and 12Å). In all cases the ROC curve of the filter is above that of XXstout, indicating an improvement. The area under the ROC curves grows narrowly for near-diagonal contacts, from 0.410 to 0.420 for 12Å and from 0.541 to 0.559 for 8Å, more consistently for long-range contacts, from 0.173 to 0.208 for 12Å and from 0.131 to 0.151 for 8Å. A typical example of prediction by XXStout and NN-Filter is shown in figures 11 and 12.

### 3.2. *Standard quality measures*

Based on ROC curves on the validation set $VS321$ (not shown) we determine breakeven points for all networks, then compare performances of all systems at breakeven. Breakeven points for 8Å maps are 0.34 for the region beside the diagonal, and 0.14 for long-range contacts. For 12Å maps breakeven points are, respectively, 0.42 and 0.08.

We measure all performances on the $TES481$ data set. The $F1$ measure for NN-Filter is an improvement over XXstout's both for 12Å and 8Å maps, and for all bands of sequence separation considered. For long range contacts (sequence separation of 24 or greater) $F1$ grows from 0.286 for XXstout to 0.316 for NN-Filter for 12Å maps and from 0.097 to 0.168 for 8Å maps.

In all cases the improvements come from a small reduction in precision P, and a much greater increase in recall R. In the case of contacts between residues with a sequence separation of 24 or greater and 8Å maps P decreases only from 23% to 21.9%, while R more than doubles from 5.7% to 13.7%. Thus NN-Filter is able to guess more than twice as many long range contacts than XXstout, with only a minor loss of accuracy.

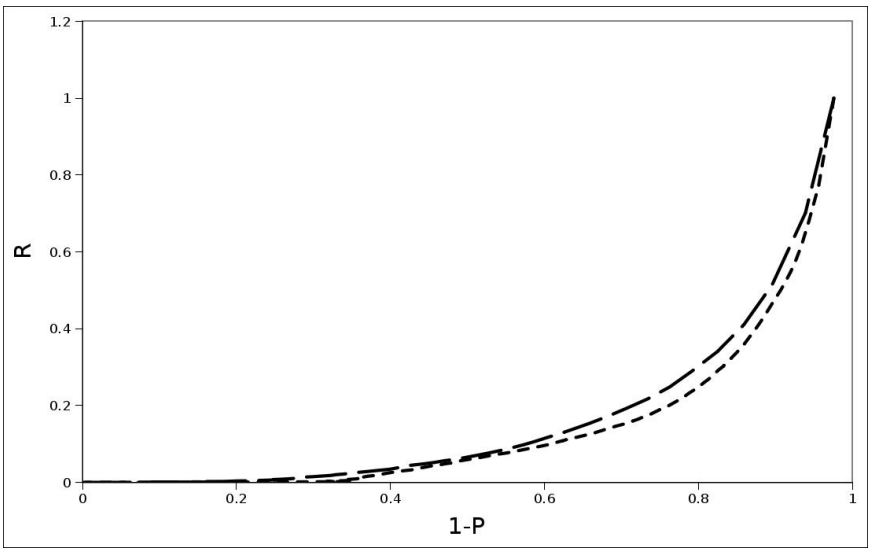


Fig. 1. ROC Curve for 8Å maps, positions away from the diagonal ($|i-j| > 8$). Long dashes=NN-Fiter. Short-dashes=XXStout. The AUC (area under the curve) is 0.131 for XXStout and 0.151 for NN-Filter.

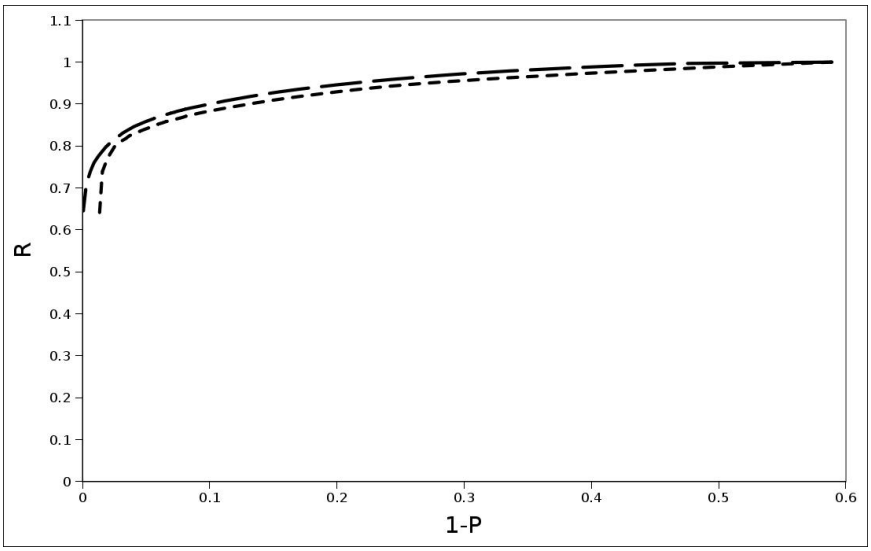


Fig. 2. ROC Curve for 8Å maps, positions beside the diagonal ($|i-j| \leq 8$). Long dashes=NN-Fiter. Short-dashes=XXStout. The AUC (area under the curve) is 0.541 for XXStout and 0.559 for NN-Filter.

### 3.3. *Physicality measures*

Figures 5, 6, 7, 8, 9, 10 show the distributions of the various physicality measures (NC, NCIJ and CO) for both 8Å and 12Å maps.

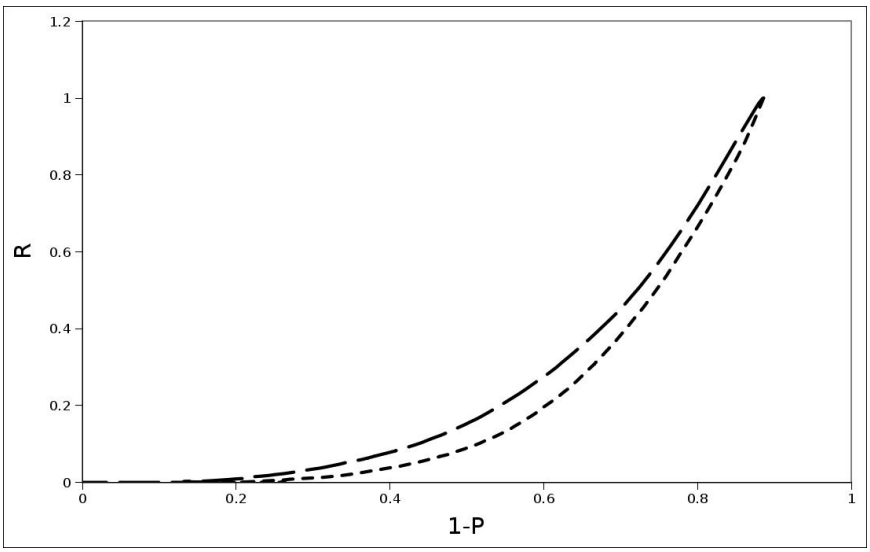


Fig. 3. ROC Curve for 12Å maps, positions away from the diagonal ($|i-j| > 12$). Long dashes=NN-Fiter. Short-dashes=XXStout. The AUC (area under the curve) is 0.173 for XXStout and 0.208 for NN-Filter.

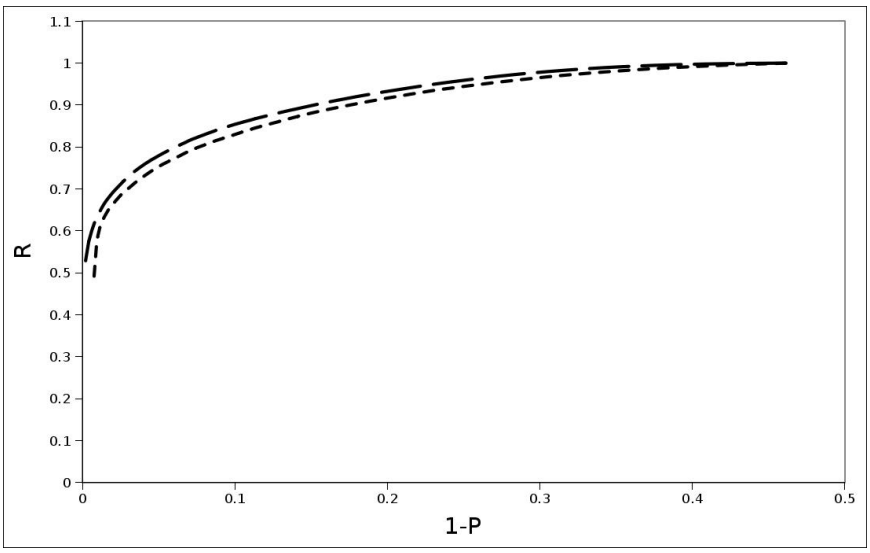


Fig. 4. ROC Curve for 12Å maps, positions away from the diagonal ($|i-j| \leq 12$). Long dashes=NN-Fiter. Short-dashes=XXStout. The AUC (area under the curve) is 0.410 for XXStout and 0.420 for NN-Filter.

To gauge the similarity between distributions, we ran Kolmogorov-Smirnov tests for XXStout vs. Native and NN-Filter vs. Native for NC, NCIJ and CO. The results,

| | XXStout | | | NN-Filter | | |
|---|---|---|---|---|---|---|
| | $F1$ | $P$ | $R$ | $F1$ | $P$ | $R$ |
| $\|i-j\| \geq 6$ | 0.396 | 0.386 | 0.407 | 0.411 | 0.336 | 0.527 |
| $\|i-j\| \geq 12$ | 0.323 | 0.313 | 0.334 | 0.353 | 0.278 | 0.484 |
| $\|i-j\| \geq 24$ | 0.286 | 0.275 | 0.298 | 0.316 | 0.254 | 0.417 |
| $\|i-j\| > 12$ | 0.319 | 0.309 | 0.330 | 0.352 | 0.275 | 0.488 |
| $\|i-j\| \leq 12$ | 0.858 | 0.857 | 0.858 | 0.876 | 0.875 | 0.876 |

Table 2. Performances for XXstout and NN-Filter on the $TES481$ data set for different sequence separations. 12Å maps.

| | XXStout | | | NN-Filter | | |
|---|---|---|---|---|---|---|
| | $F1$ | $P$ | $R$ | $F1$ | $P$ | $R$ |
| $\|i-j\| \geq 6$ | 0.213 | 0.354 | 0.153 | 0.271 | 0.277 | 0.265 |
| $\|i-j\| \geq 12$ | 0.159 | 0.315 | 0.107 | 0.235 | 0.247 | 0.224 |
| $\|i-j\| \geq 24$ | 0.097 | 0.230 | 0.057 | 0.168 | 0.219 | 0.137 |
| $\|i-j\| > 8$ | 0.186 | 0.335 | 0.129 | 0.254 | 0.256 | 0.252 |
| $\|i-j\| \leq 8$ | 0.889 | 0.888 | 0.890 | 0.899 | 0.891 | 0.906 |

Table 3. Performances for XXstout and NN-Filter on the $TES481$ data set for different sequence separations. 8Å maps.

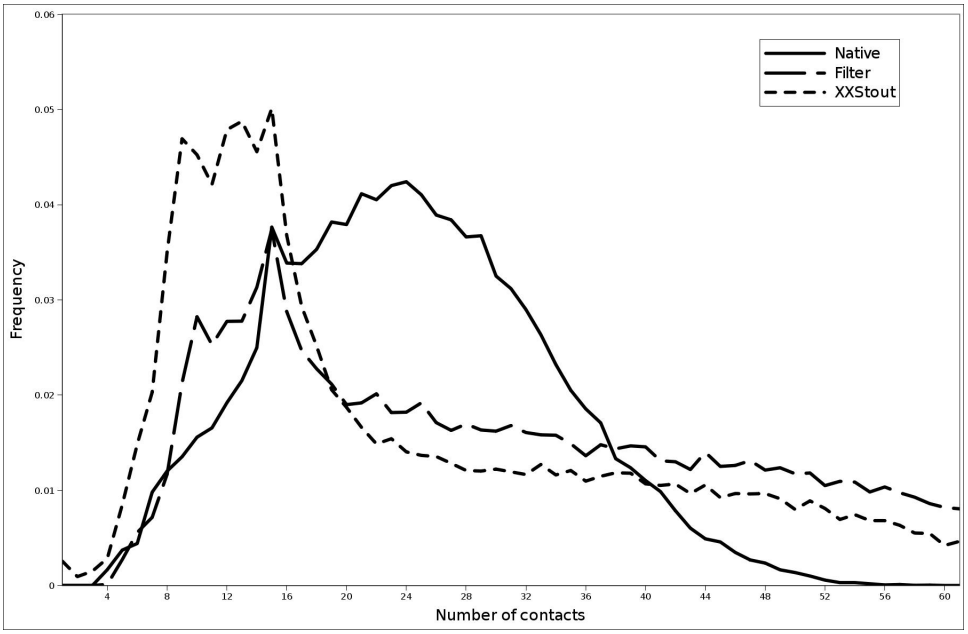


Fig. 5. Number of contacts per amino acid (NC) in the $TES481$ dataset. 12Å maps. The solid line represents the native distribution, the dashed line NN-Filter, and the dotted line XXStout.

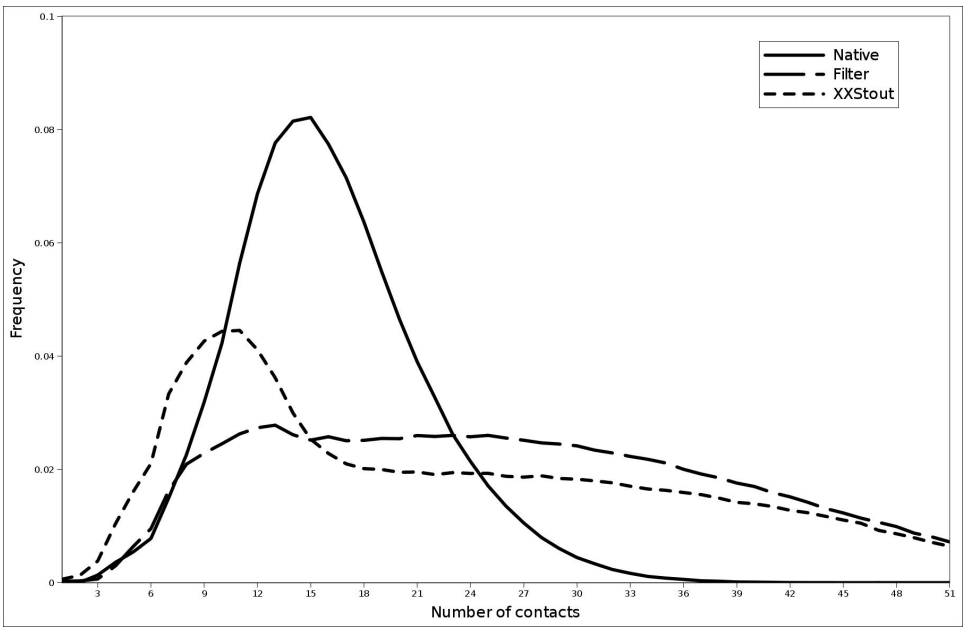


Fig. 6. Number of amino acids in contact with both AAi and AAj (NCIJ) in the $TES481$ dataset. 12Å maps. The solid line represents the native distribution, the dashed line NN-Filter, and the dotted line XXStout.

| | NC | NCIJ | CO |
|---|---|---|---|
| XXStout | 0.3315 | 0.2528 | 0.1849 |
| NN-Filter | 0.3233 | 0.3111 | 0.0933 |

Table 4. Comparison of distributions of XXStout vs. native maps and NN-Filter vs. native maps, 12Å maps. Kolmogorov-Smirnov test.

reported in Tables 4 and 5, suggest that NN-Filter is closer to the native distribution for the case of Contact Order, slightly further for NCIJ and NC at 8Å, and roughly the same for the other quantities. A visual inspection of the distributions suggests that at least part of this effect might be an artifact of the test. For instance NN-Filter NC distributions (Figs 5 and 8) "look" more similar in shape to the native ones than XXStout's, in that they are closer to the native one for nearly all NC values for which the native distribution is non-zero (for up to about 40 contacts for 12Å and just over 15 contacts for 8Å). Beyond these thresholds the native distributions become very close to zero while NN-Filter's ones are slightly higher than XXStout's, which causes a significant offset of the cumulative distributions. In practice, the slight overprediction of residues on the tail of the distributions offsets NN-Filter's better match with the bulk of the native distribution.

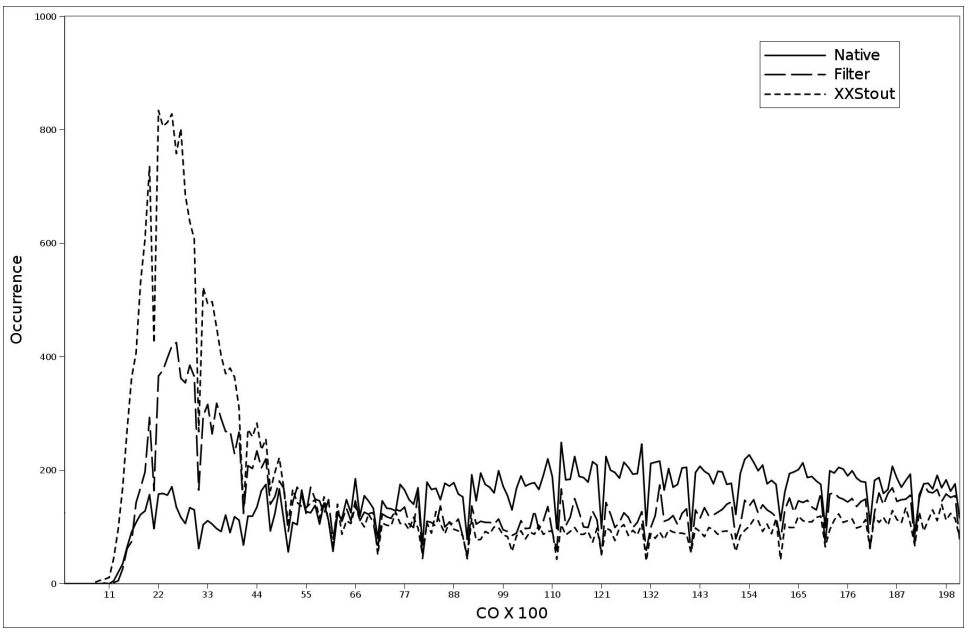


Fig. 7. Residue CO distributions in the $TES481$ dataset. 12Å maps. The solid line represents the native distribution, the dashed line NN-Filter, and the dotted line XXStout.

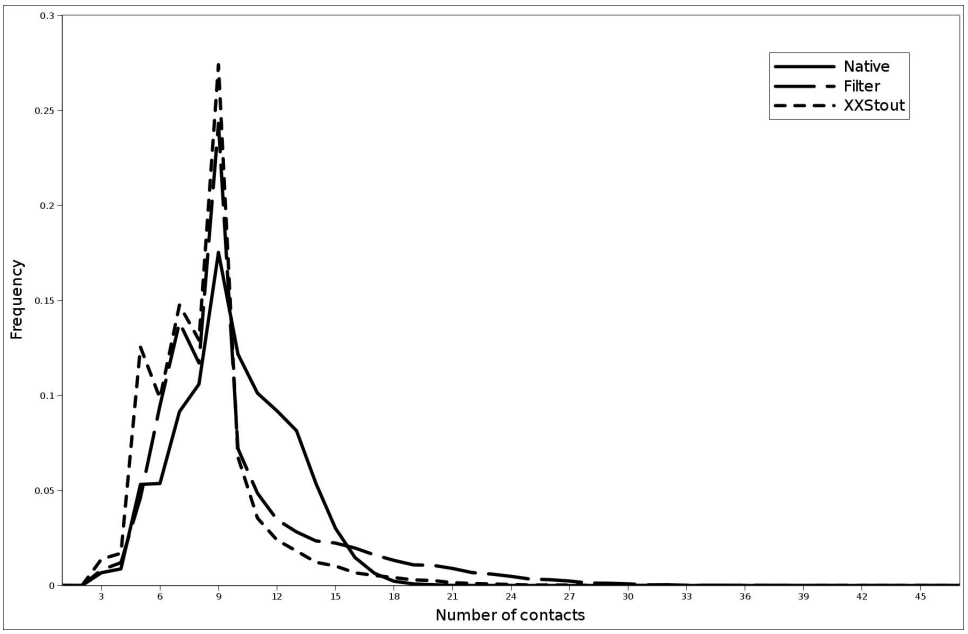


Fig. 8. Number of contacts per amino acid (NC) in the $TES481$ dataset. 8Å maps. The solid line represents the native distribution, the dashed line NN-Filter, and the dotted line XXStout.

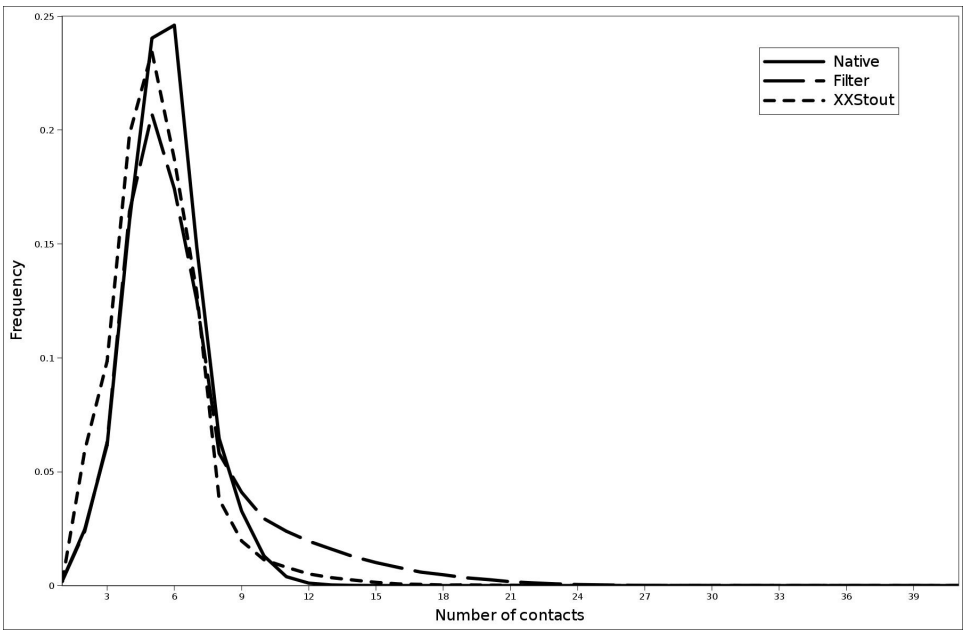


Fig. 9. Number of amino acids in contact with both AAi and AAj in the 481 dataset. 8Å maps. The solid line represents the native distribution, the dashed line NN-Filter, and the dotted line XXStout.

| | NC | NCIJ | CO |
|---|---|---|---|
| XXStout | 0.2905 | 0.2292 | 0.218 |
| NN-Filter | 0.3373 | 0.2576 | 0.1669 |

Table 5. Comparison of distributions of XXStout vs. native maps and NN-Filter vs. native maps, 8Å maps. Kolmogorov-Smirnov test.

#### 3.3.1. *Total number of contacts*

The total number of contacts predicted by XXStout and NN-Filter, vs. the number of contacts in the native maps, are reported in Table 6. While for 8Å maps NN-Filter predicts a number of contacts that is much closer to the correct one than XXStout's, this is actually reversed for 12Å maps, where NN-Filter over-predicts contacts compared to XXStout.

### 3.4. *3D reconstructions*

We also test whether improved, filtered maps yield more accurate reconstructions of protein structures. The reconstruction procedure we adopt is similar to that in [1], and is described in detail in [10] and [11]. Briefly, proteins are represented simply as $C_\alpha$ traces and a stochastic search (simulated annealing) of the space of protein

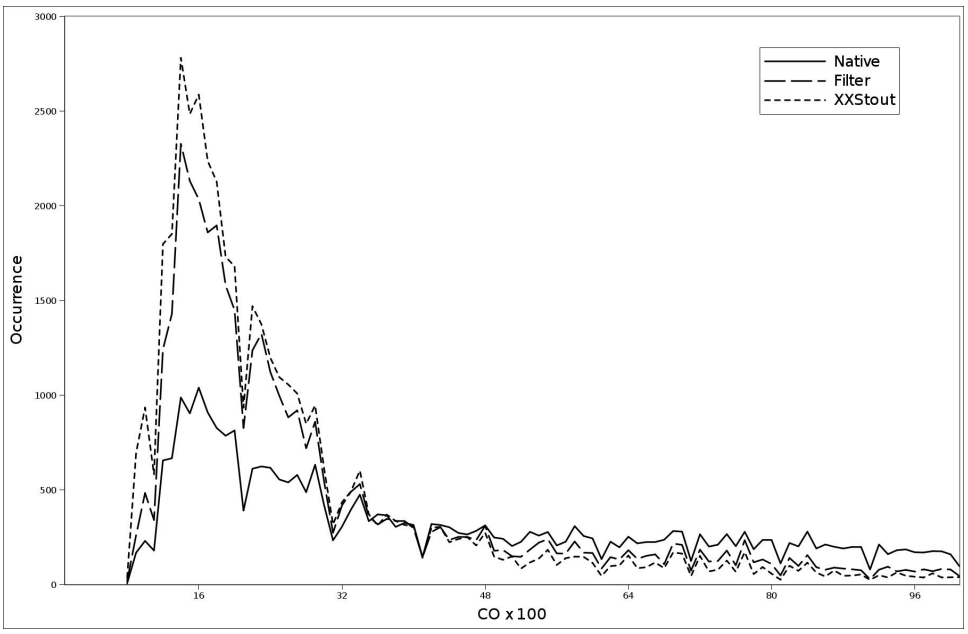


Fig. 10. Residue CO distributions in the $TES481$ dataset. 8Å maps. The solid line represents the native distribution, the dashed line NN-Filter, and the dotted line XXStout.

| | 12Å | 8Å |
|---|---|---|
| Native | 573126 | 216719 |
| XXStout | 591856 | 178749 |
| NN-Filter | 786035 | 218260 |

Table 6. Number of true (Native) and predicted contacts by XXStout and NN-Filter, 12Å and 8Å.

| | TM | GDT | RMSD |
|---|---|---|---|
| XXStout | 0.261 | 0.236 | 14.3 |
| NN-Filter | 0.260 | 0.232 | 13.8 |

Table 7. Reconstruction quality on $TES258$ for XXstout vs. NN-Filter 12Å maps.

conformations is operated, starting from a random initial structure and trying to minimise a simple cost that favours the implementation of a contact map. We reconstruct all proteins in the $TES258$ set, running 10 reconstructions for each protein, and adopting as contact maps 12Å maps from XXstout and NN-Filter. Table 7 reports root mean square distances (RMSD), GDT and TM score[15], averaged over all reconstructions, between reconstructed and native structures.

| | Distill | | | NN-Filter | | |
|---|---|---|---|---|---|---|
| $\|i-j\|$ | mF1 | mP | mR | mF1 | mP | mR |
| $\|i-j\| \geq 6$ | 0.0992 | 0.1715 | 0.0698 | 0.1320 | 0.1703 | 0.1077 |
| $\|i-j\| \geq 12$ | 0.0812 | 0.1469 | 0.0561 | 0.1182 | 0.1584 | 0.0942 |
| $\|i-j\| \geq 24$ | 0.0536 | 0.1039 | 0.0361 | 0.0831 | 0.1316 | 0.0608 |
| $\|i-j\| > 8$ | 0.0883 | 0.1563 | 0.0615 | 0.1244 | 0.1609 | 0.1014 |
| $\|i-j\| \leq 8$ | 0.8364 | 0.8236 | 0.8496 | 0.8339 | 0.8137 | 0.8551 |

Table 8. CASP7 results for XXstout (Distill) and NN-Filter. 8Å maps. The results are micro-averages, i.e. averaged over all pairs of residues in the set.

No clear conclusion can be reached from the results. Although filtered maps are an improvement over XXstout predictions, they still do not seem to contain enough information to reliably predict the structure. At close to 14Å RMSD to the native structure, predictions remain poor, at least through the simple reconstruction procedure we adopt here, and without resorting to further sources of information other than contact maps.

### 3.5. *CASP7 results*

We tested both XXstout (as part of the Distill predictor) and NN-Filter at the CASP7 competition. As NN-Filter became available during the prediction season and after some of the targets has expired (i.e. no further predictions were accepted on them) we submitted predictions only for 63 of the 95 targets (68 domains). The results in tables 8 and 9 refer to these 66 targets. The results are for a contact threshold of 8Å, as per CASP7 regulations. The first table reports micro-averaged results, meaning that precision and recall are averaged over the whole set (each pair of residues counts the same), while the second table reports macro-averages, i.e. precision and recall are computed for each protein and then averaged over all proteins (in this case each protein counts the same, but pairs of residues in long proteins count less than pairs in short proteins).

The tables broadly confirm the results on $TES481$, with NN-Filter improving over "raw" XXstout predictions on all sequence separation bands, with the exception of the area just beside the diagonal (sequence separation of at most 8), on which the two predictors are substantially undistinguishable, and quite accurate (F1 over 82%). For contacts with a sequence separation greater than 23 residues F1 grows from 5.4% to 8.3% (4.5% to 6.4% for macro-averages), and from 8.1% to 11.8% (7.7% to 11% for macro-averages) for sequence separation greater than 11.

In 10 we report the results for Distill and NN-Filter compared with other CASP7 predictors of contact maps. The results for XXStout (Distill) and NN-Filter differ from those in Tables 8 and 9 because proteins here are split in domains, as assessed by CASP. The table confirms NN-Filter's improvements over XXStout.

| | Distill | | | NN-Filter | | |
|---|---|---|---|---|---|---|
| $\|i-j\|$ | F1 | P | R | F1 | P | R |
| $\|i-j\| \geq 6$ | 0.0990 | 0.1712 | 0.0763 | 0.1229 | 0.1503 | 0.1187 |
| $\|i-j\| \geq 12$ | 0.0773 | 0.1354 | 0.0579 | 0.1098 | 0.1382 | 0.1057 |
| $\|i-j\| \geq 24$ | 0.0446 | 0.0891 | 0.0332 | 0.0640 | 0.0858 | 0.0599 |
| $\|i-j\| > 8$ | 0.0871 | 0.1463 | 0.0665 | 0.1159 | 0.1410 | 0.1148 |
| $\|i-j\| \leq 8$ | 0.8272 | 0.8083 | 0.8542 | 0.8252 | 0.8037 | 0.8558 |

Table 9. CASP7 results for XXstout (Distill) and NN-Filter. 8Å maps. The results are macro-averages, i.e. averaged over all proteins in the set.

| $\|i-j\|$ | $\geq 12$ | | | $\geq 24$ | | |
|---|---|---|---|---|---|---|
| Server | P | R | F1 | P | R | F1 |
| *Gadjapairings* | 0.148 | 0.486 | 0.227 | 0.136 | 0.502 | 0.214 |
| Betapro | 0.269 | 0.150 | 0.193 | 0.230 | 0.120 | 0.158 |
| SVMcon | 0.401 | 0.124 | 0.189 | 0.330 | 0.104 | 0.158 |
| NN-Filter | 0.145 | 0.257 | 0.185 | 0.087 | 0.199 | 0.121 |
| *PROFcon* | 0.184 | 0.125 | 0.149 | 0.123 | 0.094 | 0.107 |
| SAM_T06 | 0.382 | 0.077 | 0.128 | 0.353 | 0.073 | 0.121 |
| Distill | 0.088 | 0.269 | 0.132 | 0.054 | 0.189 | 0.084 |
| Possum | 0.086 | 0.180 | 0.117 | 0.058 | 0.164 | 0.085 |
| GPCpred | 0.064 | 0.119 | 0.083 | 0.034 | 0.131 | 0.054 |

Table 10. CASP7 results for XXstout (Distill) and NN-Filter vs. other CASP predictors. 8Å maps. The results are macro-averages, i.e. averaged over all proteins in the set, and sorted by descending F1. Results for 68 domains (63 targets) submitted by NN-Filter. Proteins are split into domains as assessed at CASP. All other predictors tested on the same 68 domains except: Gadjapairings (48 domains); PROFcon (66 domains).

## 4. Conclusions

One of the main problems with residue contact map predictors is the fact that they generally fail to incorporate global information about a protein. As a result, often, predicted contact maps are inherently local, in that their predictions do not respect some basic rules about the number of residue contacts allowed for an amino acid, its contact order, etc. In this paper we proposed a simple attempt to improve predicted contact maps by building a filter that has access not only to local properties of the maps, but also to a number of relevant global quantities. Somewhat surpisingly, it is unclear whether the filter improves the overall "physicality" of the maps (although this might be partly an artifact of the test we used), with only residue contact orders clearly becoming closer to native. Nonetheless, our results show that the filter greatly enhances the quality of the map, especially for long-range contacts,

Fig. 11. Example prediction for protein 1HRUA. Left side: bottom-left true map, top-right XXStout predicted 8Å map. Right side: bottom-left true map, top-right NN-Filtered 8Å map. In both cases the region beside the diagonal is predicted fairly well, while far from the diagonal, interactions between secondary structure elements are poorly predicted, especially interactions between strands that are not contiguous in the sequence. Beside the diagonal there is a small improvement by NN-Filter ($\Delta P = 1.4\%$, $\Delta R = 1.5\%$). Far from the diagonal NN-Filter predicts slightly better than XXStout, with interactions between secondary structure elements still poorly predicted, but some spurious interactions removed.

that are both harder to predict, and more informative to determine the overall topology of a protein structure. These results are confirmed by the latest CASP7 competition, in which the filtered architecture clearly outperformed the first-stage predictor. Importantly, since the filtering stage is entirely modular, we believe that our approach is likely to be beneficial to any contact map predictor.

We are currently extending this work to deal with distance maps, and multi-class distance maps, which are more informative than binary contact maps, and with maps incorporating homology to structures in the PDB, which are greatly more accurate than *ab initio* predicted ones. In both cases, it is likely that improvements in map quality will also translate into improvements in 3D structure prediction.

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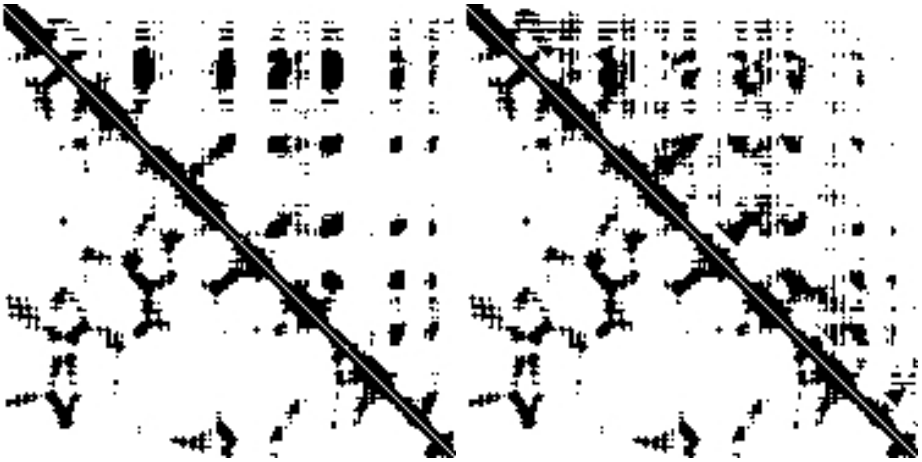

Fig. 12. 1HRUA. Left side: bottom-left true map, top-right XXStout predicted 12Å map. Right side: bottom-left true map, top-right NN-Filtered 12Å map. Simlarly to the 8Å case positions beside the diagonal are fairly well predicted, but interactions far from the diagonal are not. As for 8Å case there is a small improvement by NN-Filter near the diagonal. Far from the diagonal, the dense patches of contacts erronously predicted by XXStout have thinned out, although some isolated wrongly predicted contacts have appeared.

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